

1. **Final answer:** $\pi(2 - \sqrt{2}) \ln 2$.

Use spherical coordinates

$$x = \rho \sin \varphi \cos \theta, \quad y = \rho \sin \varphi \sin \theta, \quad z = \rho \cos \varphi,$$

where φ is measured from the positive z -axis. The sphere inequalities give $1 \leq \rho \leq 2$. Since $\rho > 0$, the cone inequality becomes $\rho \cos \varphi \geq \rho \sin \varphi$, so $0 \leq \varphi \leq \pi/4$. There is no azimuthal restriction. The Cartesian conditions and the spherical bounds are

$$\begin{cases} 1 \leq x^2 + y^2 + z^2 \leq 4, \\ z \geq \sqrt{x^2 + y^2} \end{cases} \implies \begin{cases} 0 \leq \theta \leq 2\pi, \\ 0 \leq \varphi \leq \pi/4, \\ 1 \leq \rho \leq 2. \end{cases}$$

The integrand is ρ^{-3} , and the volume element is $dV = \rho^2 \sin \varphi d\rho d\varphi d\theta$. Consequently,

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^{\pi/4} \int_1^2 \frac{\sin \varphi}{\rho} d\rho d\varphi d\theta \\ &= 2\pi [-\cos \varphi]_0^{\pi/4} [\ln \rho]_1^2 = 2\pi \left(1 - \frac{1}{\sqrt{2}}\right) \ln 2 = \pi(2 - \sqrt{2}) \ln 2. \end{aligned}$$

The result is positive, as expected; the inner sphere excludes the singularity at the origin.

Full credit: any exact expression equal to $\pi(2 - \sqrt{2}) \ln 2$, including $2\pi(1 - 1/\sqrt{2}) \ln 2$. An unevaluated integral alone is incomplete. Equivalent full-turn θ -intervals give the same setup.

2. **Final answer:** $S = 4\pi$.

For the fly's path,

$$\begin{aligned} \vec{r}(t) &= (\cos t, \sin t, 2 + \cos t + \sin t), \\ \vec{r}'(t) &= (-\sin t, \cos t, \cos t - \sin t), \\ \|\vec{r}'(t)\| &= \sqrt{\sin^2 t + \cos^2 t + (\cos t - \sin t)^2} = \sqrt{1 + (\cos t - \sin t)^2}. \end{aligned}$$

The scent intensity along the path is

$$\sigma(\vec{r}(t)) = \sqrt{1 + (\cos t - \sin t)^2} = \|\vec{r}'(t)\|.$$

For a scalar line integral, multiply the scalar field by the speed: $ds = \|\vec{r}'(t)\| dt$, not just dt . Thus

$$\begin{aligned} S &= \int_0^{2\pi} \sigma(\vec{r}(t)) \|\vec{r}'(t)\| dt \\ &= \int_0^{2\pi} (1 + (\cos t - \sin t)^2) dt = \int_0^{2\pi} (2 - \sin 2t) dt \\ &= [2t + \frac{1}{2} \cos 2t]_0^{2\pi} = 4\pi. \end{aligned}$$

This is scent intensity weighted by distance traveled, not a time integral of scent intensity. Reversing the flight direction would not change this scalar line integral.

Full credit: any exact expression equal to 4π . An unevaluated integral alone is incomplete.

3. **Final answer:** -2.

The path follows increasing t from $(\pi/2, \pi^2/4)$ to (π, π^2) . Substitute $x = t$, $y = t^2$ into both components of the field:

$$\vec{F}(\vec{r}(t)) = \left(\frac{t^2 \cos t}{t^2}, -\frac{\sin t}{2t} \right) = \left(\cos t, -\frac{\sin t}{2t} \right), \quad \vec{r}'(t) = (1, 2t).$$

Here $t \geq \pi/2 > 0$, so the denominators never vanish. The vector line integral uses the dot product with $\vec{r}'(t)$:

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_{\pi/2}^{\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\ &= \int_{\pi/2}^{\pi} (\cos t - \sin t) dt = [\sin t + \cos t]_{\pi/2}^{\pi} = -1 - 1 = -2. \end{aligned}$$

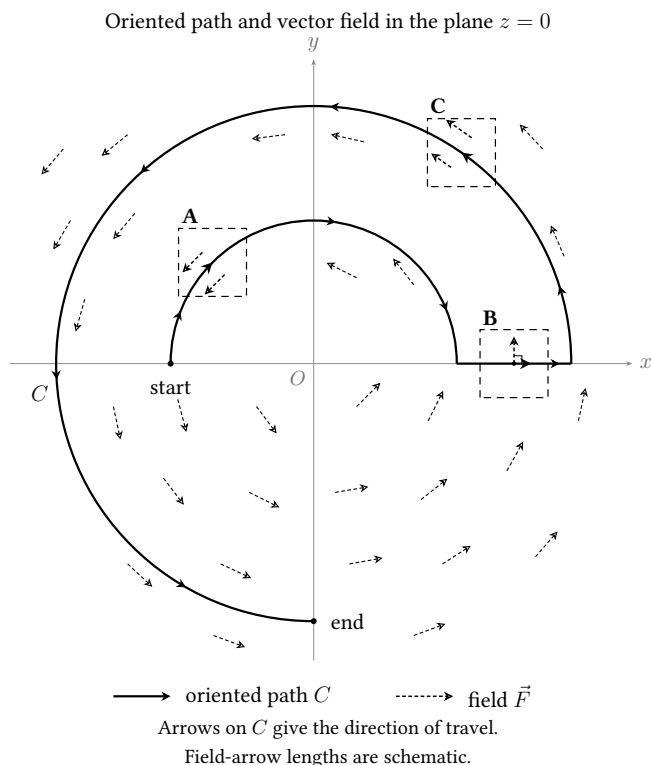
Do not insert an additional speed factor: $d\vec{r} = \vec{r}'(t) dt$. Reversing the orientation would give $+2$, which is not the requested integral.

Full credit: any exact expression equal to -2 . An unevaluated integral alone is incomplete.

4. **Final answers:**

Square	Field relative to the oriented tangent	Contribution
A	Opposite direction	Negative ($-$)
B	Perpendicular	Zero (0)
C	Same direction	Positive ($+$)

For the oriented unit tangent $\vec{T}$, the infinitesimal contribution is $\vec{F} \cdot \vec{T} ds$. The path's filled arrowheads determine $\vec{T}$; the thinner dashed arrows represent $\vec{F}$. Opposite, perpendicular, and matching directions give negative, zero, and positive dot products, respectively. These alignments hold throughout each highlighted piece, so the signs also describe the integrals over those pieces.



Instructor check: the hidden field is $\vec{F}(x, y, z) = (-y, x, 0)$. Its dot product with the oriented unit tangent is -2 on the inner clockwise arc, 0 on the outward radial segment, and $18/5$ on the outer counterclockwise arc. Arrow lengths in the drawing are schematic.

Full credit: A negative, B zero, C positive, accepting either words or the symbols $-$, 0 , $+$ in the corresponding entries. Treat the three entries as equally weighted components; no written explanation is required. The question does not ask for the sign over the entire path.